Engineering Structures 46 (2013) 218–233

Contents lists available at SciVerse ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct

Pushover experiment and analysis of a full scale non-seismically detailed RC structure

Akanshu Sharma [a,*], G.R. Reddy [a], K.K. Vaze [a], R. Eligehausen [b]

[a] Reactor Safety Division, Bhabha Atomic Research Centre, Mumbai, India
[b] Institute for Construction Materials, University of Stuttgart, Germany

ARTICLE INFO

Article history:
Received 7 February 2011
Revised 6 August 2012
Accepted 7 August 2012
Available online 13 September 2012

Keywords:
Pushover
Full-scale experiments
RC frames
Non-seismic detailing
Failure patterns
Modeling techniques

ABSTRACT

The paper presents experimental and numerical work carried out on a full-scale four storey reinforced concrete (RC) structure for seismic assessment by pushover method. For practicality, a portion of an existing structure having certain eccentricities was replicated for the experimental setup. The structure was detailed as per non-seismic reinforcement detailing norms of Indian Standards. The experiment was carried out as a round robin exercise, in which various institutes in India participated and presented pre-test results in the form of pushover curves. A large variation in the pre-test results highlighted that the result of a pushover analysis is highly sensitive to the adopted modeling techniques. This paper reports the details and results of the experiment and focuses on the need of modeling various structural nonlinearities to obtain realistic results. The results of pre-test analysis by various research groups, in which the emphasis was given on modeling issues as well as a more efficient post-test numerical procedure is also presented and compared. It is shown that a basic pushover analysis considering only flexural failures may not be able to achieve a realistic simulation, thus it is mandatory to develop relatively simple, yet effective models to consider more complex phenomena such as joint shear failures, to achieve realistic predictions.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

As more and more emphasis is given on non-linear analysis of RC framed structures subjected to earthquake excitations, the research and developments on non-linear static (pushover) analysis as well as nonlinear dynamic (time history) analysis are in the forefront. Due to prohibitive computational cost required to perform a complete nonlinear dynamic analysis, researchers and designers all over the world are showing keen interest in nonlinear static pushover analysis. This is proven by the fact that codes such as ATC 40 [1], FEMA 273 [2] followed by FEMA 356 [3] and more recently FEMA 440 [4] have given detailed guidelines to perform the nonlinear static pushover analysis and to use it to assess the performance of structures under a given earthquake scenario. The post processing procedures recommended to determine the performance of the structure against a given earthquake vary significantly in codes, with ATC 40 [1] recommending capacity spectrum method, FEMA 356 [3] describing displacement coefficient method and FEMA 440 [4] describing improvements to both capacity spectrum and displacement coefficient methods. However, all these procedures require determination of nonlinear force–deformation curves that are generated from pushover analysis. This simplifies the structural model and provides useful information about the likely non-linear behavior of the structure. Therefore, it is evident that a vital step towards good seismic performance estimation of the structure is reliable and accurate determination of force–deformation curve, generally known as "pushover curve" or "capacity curve".

In order to capture the complete picture of the nonlinear behavior of the frame structure, it is required to model various nonlinearities, such as flexural, shear, axial and torsional behavior (with interactions) for beams and columns as well as to predict more complex behavior at the connections, e.g., joint shear failure, bond failure, etc. Especially, for the structures designed and detailed according to non-seismic detailing practice, the nonlinear behavior of the beam–column joints plays a dominant role in determining the seismic response of the structure. The modeling of inelastic behavior of the members under flexure, shear, torsion and axial forces are well documented and several models and methods are available in the literature to simulate such characteristics [5–8]. Certainly, one can find many more models and methods suggested in literature and the references mentioned here are just indicative. Typically, while analyzing a structure by pushover method, only member non-linearities are considered and the joints are assumed rigid. However, several studies have introduced models to account for inelastic shear deformation and bar bond slip [9–13].

* Corresponding author. Tel.: +91 22 25591530.
E-mail addresses: akanshu@barc.gov.in, akanshusharma@yahoo.co.in (A. Sharma).

0141-0296/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.engstruct.2012.08.006

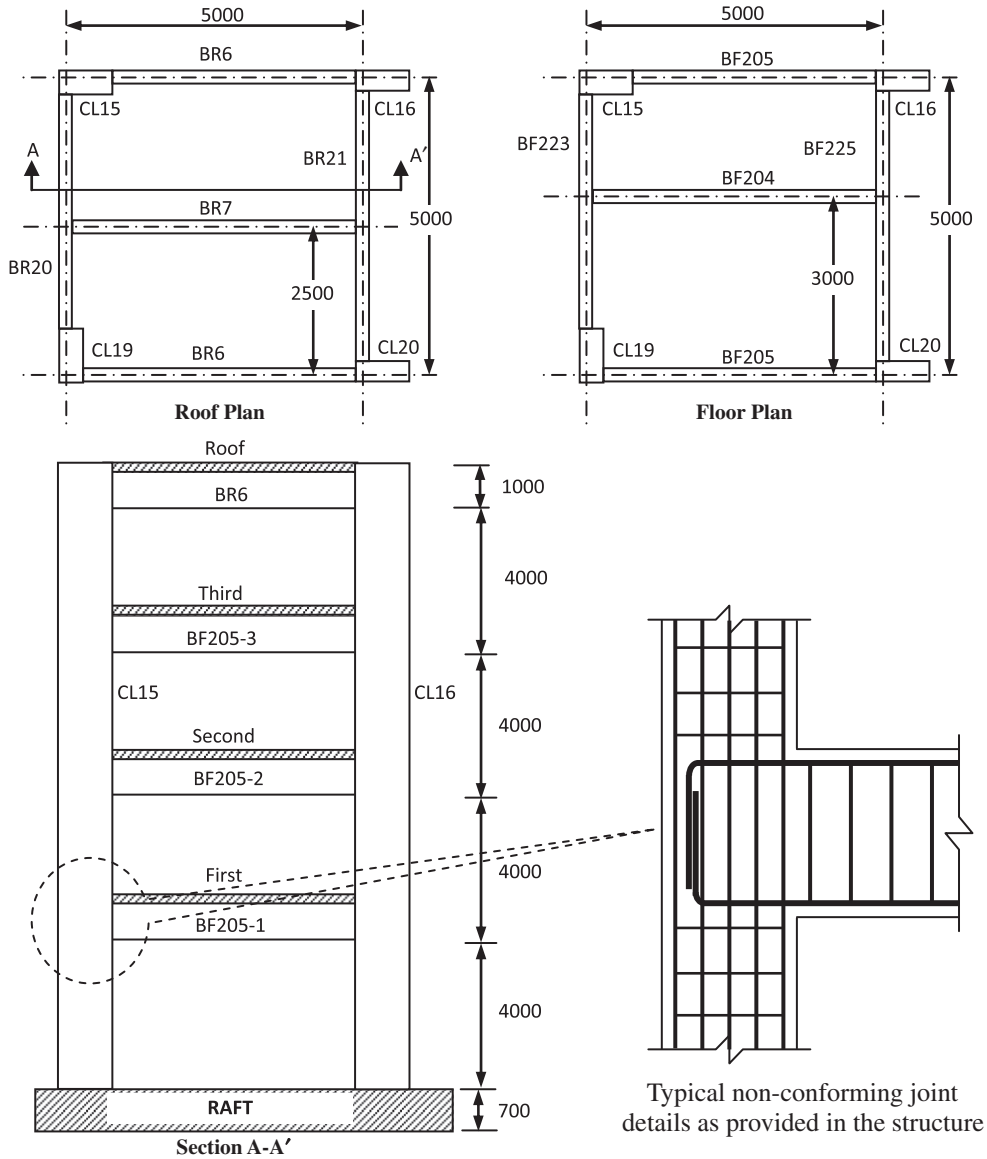


Fig. 1. Geometry of the structure.

Table 1
Details of structural members.

Beam/Column	B (mm)	D (mm)	Long. reinforcement	Trans. reinforcement
BF 204	230	1000	2–16 (Top) 3–16 (Bottom)	8–200 c/c
BF 205	230	1000	2–25 (Top) 2–25 (Bottom)	10–125 c/c
BF 223	230	1000	2–25 (Top) 2–25 + 1–16 (Bottom)	10–125 c/c
BF 225	230	1000	2–20 (Top) 2–25 (Bottom)	10–150 c/c
BR 6	230	1000	2–20 (Top) 3–20 (Bottom)	8–200 c/c
BR 7	230	600	2–16 (Top) 3–16 (Bottom)	8–120 c/c
BR 20	230	1000	2–20 (Top) 2–25 (Bottom)	8–175 c/c
BR 21	230	1000	2–20 (Top) 2–20 (Bottom)	8–175 c/c
CL 15/ CL 19 (Gr to 2 floor)	400	900	12–28	10–100 c/c
CL 15/ CL 19 (2–3 floor)	400	700	4–25 + 6–20	10–100 c/c
CL 15/ CL 19 (3–4 floor)	300	700	8–20	10–100 c/c
CL 16/ CL 20 (Gr to 2 floor)	350	900	12–25	10–100 c/c
CL 16/ CL 20 (2–4 floor)	350	900	10–20	10–100 c/c

The validation of any analysis type requires comparison of numerical results with those of the experiments. The experiments on full-scale real life type structure is the best way not only to study the behavior of the structures under lateral seismic loading, but to provide useful results that can be used to form a database to validate the analysis procedures. Efforts have been made in the past to perform tests on full-scale structures under pseudo-dynamic loads [14–16], Pushover loads [17,18] and cyclic loads

[19], but the database is not too extensive due to prohibitive cost, time and efforts involved.

In this work, one such experiment was attempted where, a 3-D full-scale structure four storey high and having one bay along both horizontal directions, was loaded under monotonically increasing lateral pushover loads. The details of design, construction, detailing, loading, instrumentation and experimental results are reported in this study. The failure patterns clearly displayed the vulnerability of RC buildings with non-conforming detailing, since they fail in undesirable failure mechanisms, such as joint shear failures.

Furthermore, the pushover results provided by several participants of the round robin exercise emphasized that the response is highly sensitive to the assumptions that have been used. The most important characteristic observed in the models of several participants was found to be "giving no consideration to joint distortion". A post-test pushover analysis of the structure was performed by the authors with certain extra considerations in modeling which were influenced by experimentally observed failure modes. The modeling approach presented here has been earlier validated and utilized by the authors [49]. The appropriate assumptions required to capture the non-linear static response of a given structure are presented. The details of the analysis are given in the sequence.

Although relatively large work has been done by various researchers to improve the predictions of demand on the structures, such as modal pushover analysis [20–25], simplified procedures using response spectrum techniques [26–28] and incremental dynamic analysis [29,30], the evaluation of structural capacity has taken a backseat, which is mainly due to the fact that due to the lack of experimental data, the results of analysis are relied upon and considered adequate. The two main objectives of this research are:

1. To contribute towards the database on experiments on full-scale RC structures under seismic loading, while providing valuable information on the overall seismic behavior of the structure, different failure modes, etc.
2. To emphasize the need to develop and suggest suitable modeling techniques, which can efficiently predict the behavior of non-conforming RC structures.

Table 2
Average compressive strength for concrete.

Location	Average Compressive Strength (MPa)
Raft	32.88
Columns 1st Floor	28.86
Beams and Slab 1st Floor	27.73
Columns 2nd Floor	33.30
Beams and Slab 2nd Floor	31.09
Columns 3rd Floor	32.24
Beams and Slab 3rd Floor	29.86
Columns 4th Floor	31.24
Beams and Slab 4th Floor	30.56

Table 3
Properties of Reinforcement.

Dia. Of bar (mm)	Yield strength (MPa)	Ult. strength (MPa)
8	456.06	604.91
10	517.81	599.94
12	539.88	620.78
16	490.96	615.02
20	488.93	614.60
25	523.37	629.49
28	498.42	622.33

2. Description of the RC frame

2.1. Geometry

A part of an existing RC office building was selected for testing. Fig. 1 shows the general geometric arrangement of the structure. The typical beam size was 230 mm × 1000 mm and the column size varied from 400 mm × 900 mm to 300 mm × 700 mm as shown in Table 1. The slab thickness was 130 mm. As seen in Fig. 1, the secondary beam (BF204) in all floors was eccentric, whereas the same for fourth floor (BR7) was in the centre of the slab. This was as provided in original structure to support the load from partition wall on 1st 2nd and 3rd floor slabs, whereas there was no such wall on the roof. Table 1 presents the longitudinal reinforcement for the beams is mentioned in the "number of bars – diameter of bars in mm (location of reinforcement in the section)", e.g., 2–16 (Top) refers to two 16 mm diameter bars located at the top of the section (to act as compression reinforcement under sagging moment). The longitudinal reinforcement for the columns is distributed uniformly along the periphery and is mentioned as "number of bars – diameter of bars in mm" format, e.g., 12–28 refers to twelve 28 mm diameter bars distributed uniformly along the perimeter of the column. The transverse reinforcement is mentioned as "diameter of stirrups/ties (in mm) – spacing of stirrups/ties (in mm)", e.g., 8–200 refers to 8 mm diameter bars as stirrups/ties spaced at a centre to centre spacing of 200 mm.

2.2. Material properties

For each floor level and for columns extending from one floor to another, six standard 150 mm cubes were tested under compressive loads and the average 28 day strength was obtained. Table 2 depicts the values of compressive strength obtained for different levels. Cold worked deformed bars with a nominal strength of 415 MPa [31] were used in construction. The properties for bars of different diameters are given in Table 3.

3. Construction of the frame

3.1. General description

There was one major change in the reinforcement detailing of the test structure as compared to the original building. The original structure was detailed following the confined (conforming) detailing practice as per IS 13920:1993 [32], whereas unconfined (non-conforming) detailing was adopted for the tested structure. The reason for such modification lies in the fact that pushover analysis is commonly used for seismic re-qualification of old buildings, which generally follow non-conforming detailing. Moreover, it is more demanding from computational point of view. In addition, since the examined frame is a substructure of a larger structure, the continuous reinforcements in the slab and beams were suitably modified. Fig. 1 shows a typical non-conforming joint detail as was constructed in the structure. The beam longitudinal reinforcement bars were extended beyond the face of the column into the joint up to a length equal to the development length for the bar as calculated by Indian code, IS 456:2000 [33].

3.2. Foundation

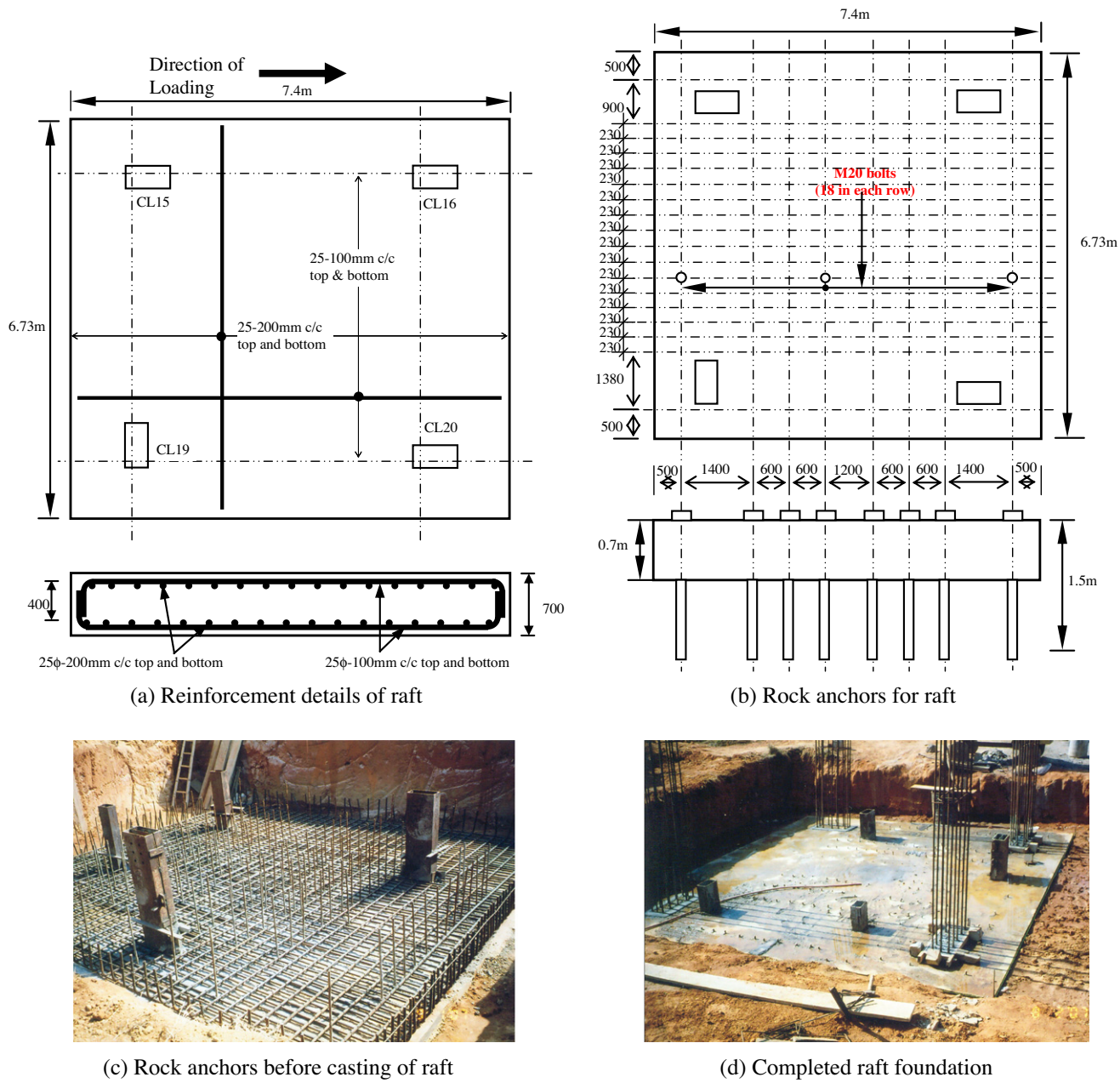
One of the major challenges during the construction of the frame was to restrict the possible rotation of the foundation of the structure. This was practically not possible if simple footings were made. Therefore, foundation for the structure was formed as a uniform raft for all columns. The substratum was found to

be hard rock and therefore, in order to avoid any possible rotation of the foundation, rock anchors were provided. In total, 144 numbers of 1.5 m long rock anchors were used with 700 mm embedment in concrete and 800 mm in rock. It is important to note that this man-made fixity has an effect on the contribution of structural modes of vibration, since in a flexibly supported system the modal mass activated by the first mode is lower than that of a corresponding fixed based system. However, since the objective of this work was to study and model the inelastic behavior of the structure, such a simplification was deemed necessary to avoid any soil-structure interaction effects, which is a separate research topic of its own.

The raft was constructed in such a way that the clear overhang of the raft is equal to 750 mm from the face of each column on both sides. Thus, the raft dimensions were 7.40 m × 6.73 m. The design details and construction of the foundation are shown in Fig. 2. Once the foundation was set, the superstructure was cast in stages as any other normal building construction with a quality control similar to the general quality control followed during the construction of normal residential buildings in India.

3.3. Loading arrangement

The load applied on the structure was maintained at a ratio of P: 2P: 3P: 4P corresponding to 1st floor: 2nd floor: 3rd floor: 4th floor, respectively, resulting in an inverted triangular load pattern. The load was applied by pulling the structure using the loading arrangement shown in Fig. 3. The load was applied remotely by means of high strength cables passing through pulleys using electro-mechanical winches controlled through a programmable logic control PLC (SCADA) system. The applied loads were continuously monitored using tension-type load cells.



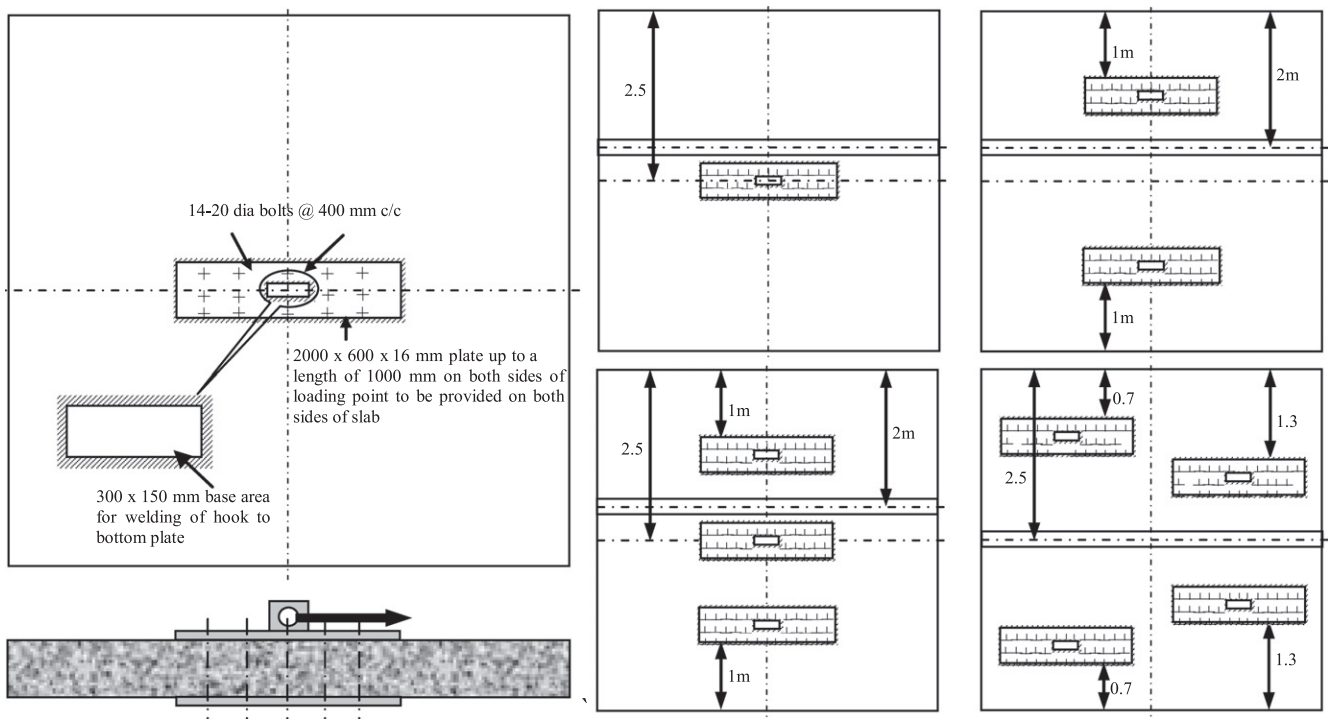
(a) Reinforcement details of raft

(b) Rock anchors for raft

(c) Rock anchors before casting of raft

(d) Completed raft foundation

Fig. 2. Details of raft foundation for the structure.



(a) Details of loading fixture (b) Arrangement at different floors

Fig. 3. Loading arrangements to apply load on the slab.

Fig. 4. Structure being tested at tower testing facility.

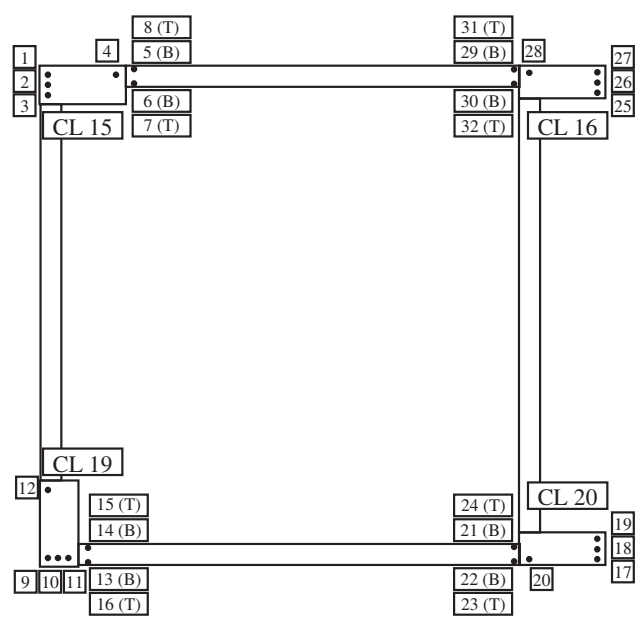


Fig. 5. Locations of strain gauges on reinforcement bars.

4. Experimental setup

4.1. Test facility

The test was conducted at tower testing facility of Central Power Research Institute, Bangalore. The facility is generally and regularly used to perform monotonic load tests on full-scale transmission line towers. The test facility is well equipped with high strength cables, pulleys, calibrated load cells, electro mechanical winches with PLC control for accurate and simultaneous load application in a predefined pattern. It has to be noted that the facility could perform the test only in load-controlled mode. This may not truly be a limitation since the pre-peak curve is generally accepted to be more accurate in the case of load-control, though a displacement control is required to capture post-peak degradation. Therefore, it would be best to perform the test under load-control in pre-peak region and under displacement control in post-peak region. However, keeping in mind the technical capabilities of

the facility and also financial and time considerations, the whole experiment was conducted in load-control mode. Fig. 4 depicts the structure being tested at the tower test facility.

4.2. Instrumentation

The instrumentation used in the experiment included:

(i) Load cells to monitor and apply the load on the structure in controlled manner.
(ii) Digital theodolites on either side of the structure (one towards CL 16 and one towards CL 20 side), to measure displacements and laser based displacement measuring devices to verify the recorded displacements.
(iii) Strain gauges on reinforcement bars to obtain strain data. Fig. 5 shows the location of strain gauges on each floor, where T and B in braces indicate the strain gauge number for top and bottom beam reinforcement, respectively.
(iv) Tilt meters for measuring member and joint rotations, which were mounted directly on the structure at the beam and column intersecting at the joint.
(v) Digital dial gauges to provide information on surface strains at the base of the columns at raft level. It was anticipated that the strains will be maximum at the base of the columns and therefore, the survival of reinforcement strain gauges or concrete surface strain gauges was doubtful. Therefore, in order to obtain average surface strains over a gauge length, at the base of the columns, dial gauge potentiometers were installed on the tension side of all the four columns.

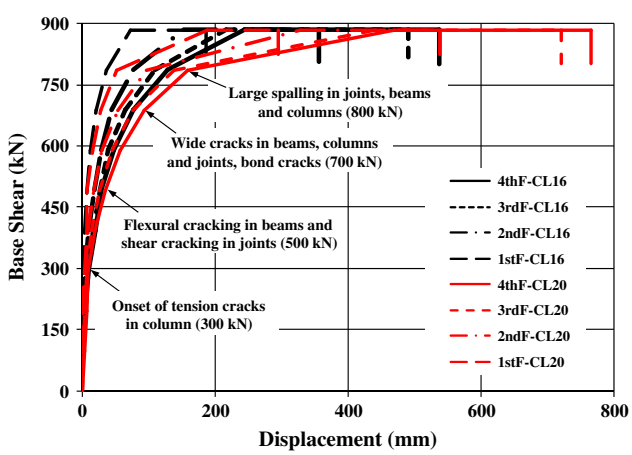


Fig. 6. Pushover curves for the structure.

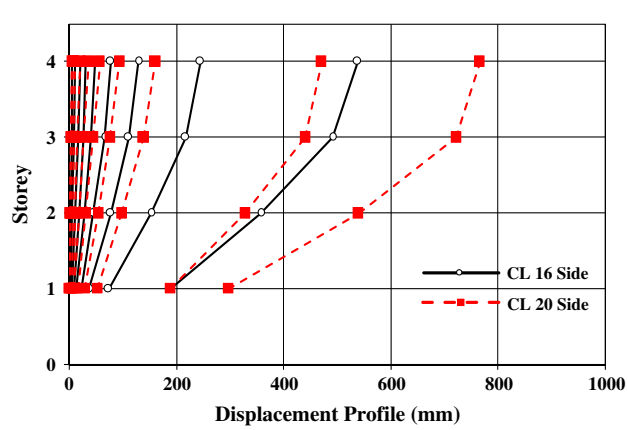


Fig. 7. Displacement pattern for increasing top drift.

4.3. Loading sequence

The loading sequence during the test was kept such that the load in the first floor was increased in the steps of 1 t (9.81 kN). Thus, the load in the second floor was incremented with the steps of 2 t (19.62 kN), that in 3rd floor in steps of 3 t (29.43 kN) and in 4th floor in steps of 4 t (39.24 kN). Thus the ratio of 1:2:3:4 is always maintained.

5. Experimental results

The pushover curves as obtained for CL16 and CL20 are shown in Fig. 6. Since the experiment was conducted under load control, the dropping part of the curves could not be obtained. Though the flat portion of Fig. 6 cannot be used for evaluation of ductility or post-yield behavior, it is still provided in order to illustrate the effect of eccentricity of the columns on the global behavior of the structure. As it can be seen in Fig. 6, the maximum displacement for CL16 side was equal to 537 mm and that on CL20 was equal to 765 mm. The considered structure is non-symmetrical in plan with one column (CL 19) section having its major axis perpendicular to the major axis of the other three columns (Fig. 1). This eccentricity and the loading arrangement design leads to a situation where the point of application of the resultant force at a particular floor does not coincide with the centre of rigidity of structure in plan. Therefore, as the lateral load is applied on the structure, the eccentricity between the point of application of resultant force and the centre of rigidity leads to storey twist in the structure [6]. Since the stiffness of the frame formed by CL 19 and CL 20 is less than that of the frame formed by CL16 and CL20 side, the storey twist results in larger displacements of the CL20 side than CL 16 side. The average top drift is equal to approximately 4% of the total height of the building.

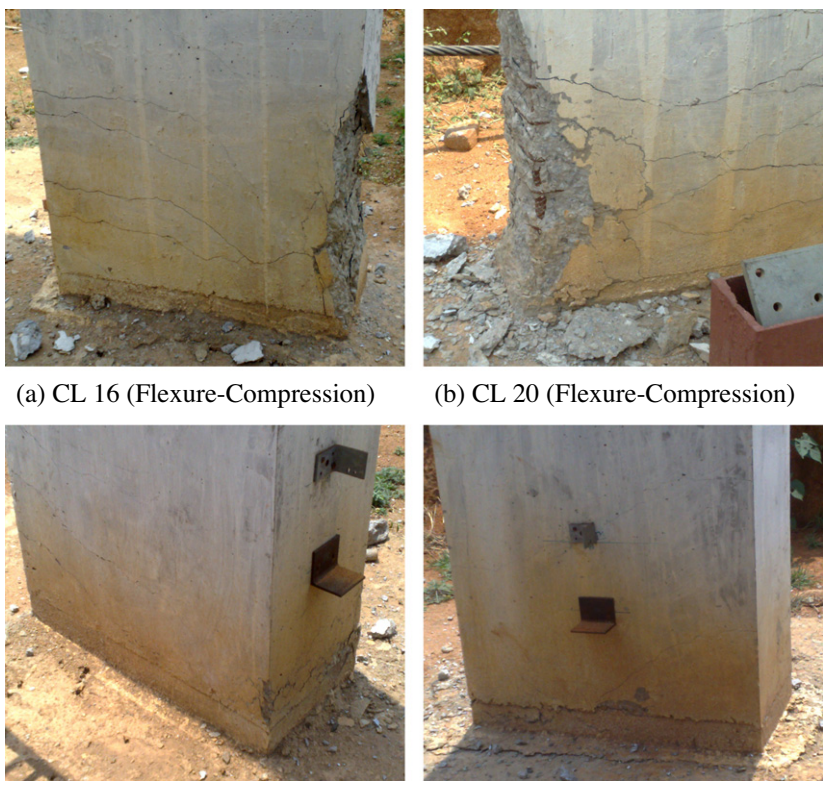
When subjected to lateral forces, the structure acts as a vertical cantilever. The resulting total horizontal force and overturning moment is transmitted at the foundation level [6]. It is evident that the structure behaved linearly up to a base shear value of approximately 300 kN. At this point the bending moments at the base of the columns caused the flexural tension cracks to appear and the structure displayed a reduced stiffness thereon (Fig. 6). The structure possesses a strong column-weak beam configuration. On further increasing the load, at a base shear value of approximately 500 kN, the cracks at the base of the columns became wider and failures at other locations, namely beams and beam-column joints began to appear. As a result the stiffness of the structure further decreased, as it can be observed in the pushover curves. Though the formation of hinges in beams after the hinge formation of the base of the columns results in a kinematically admissible mechanism [6], the failure of beam-column joints is undesirable. This is one of the prime weaknesses of non-seismically detailed structures. The joint failures are observed due to inadequate shear resistance of the core and/or poor bond behavior of bars extending into the joint, both brittle and undesirable failure modes for a structure. When the structure is subjected to cyclic loads, such failures lead to ill-formed hysteretic loops with significant pinching behavior, mainly due to slippage of reinforcing bars. Therefore, the energy absorbed by the structure due to hysteresis becomes significantly lower than that would be expected in a structure displaying only desirable beam flexural failure modes.

The failure of beam-column joints is inherently brittle and results in limited ductility, thus, degrading the seismic behavior of the overall structure as well. After reaching the base shear value of 700 kN, the joints of the structure displayed rapid degradation and the inter-storey drift increased progressively. On further increase of the lateral load, the structure displayed a very soft behav-

ior with large displacement increase for the same increase in the base shear. For a base shear of 90tn (882.90 kN), i.e., 9tn load at first floor, 18tn at second floor, 27 t at third floor and 36tn at fourth floor, the structure started undergoing increasing displacement and its resistance reduced. However, due to load control, the load decrease could not be recorded. Hence, since the load could not be increased any further, the test was stopped and the load was removed.

Though, it is rather intuitive to deduce information on the seismic capacity of non-seismically detailed structures based on the results of the experiment, it must be noted that in this experimental setup no superimposed dead loads (e.g., floor finish), no live loads, no masonry walls, etc. were included. Furthermore, the foundation of the structure was artificially fixed. These aspects prohibit a direct deduction on the seismic capacity, in terms of load or ductility, from this experiment. The objective of this experiment is to highlight the important aspects that must be considered while modeling the structure to obtain realistic predictions. Once an accurate and reliable simulation technique and numerical analysis procedure is established, it can be utilized to assess the seismic capacity of any real-life structure. Therefore, the above-discussed results were not used to draw any conclusions on the seismic capacity of structures with non-conforming detailing.

Fig. 7 shows the displacement profiles of the structure due to the applied load. Each curve corresponds to the displacement profile of a particular load step. Initially, when the structure was loaded, it behaved fairly linearly till the third load step corresponding to a base shear of 300 kN. As the lateral load on the structure



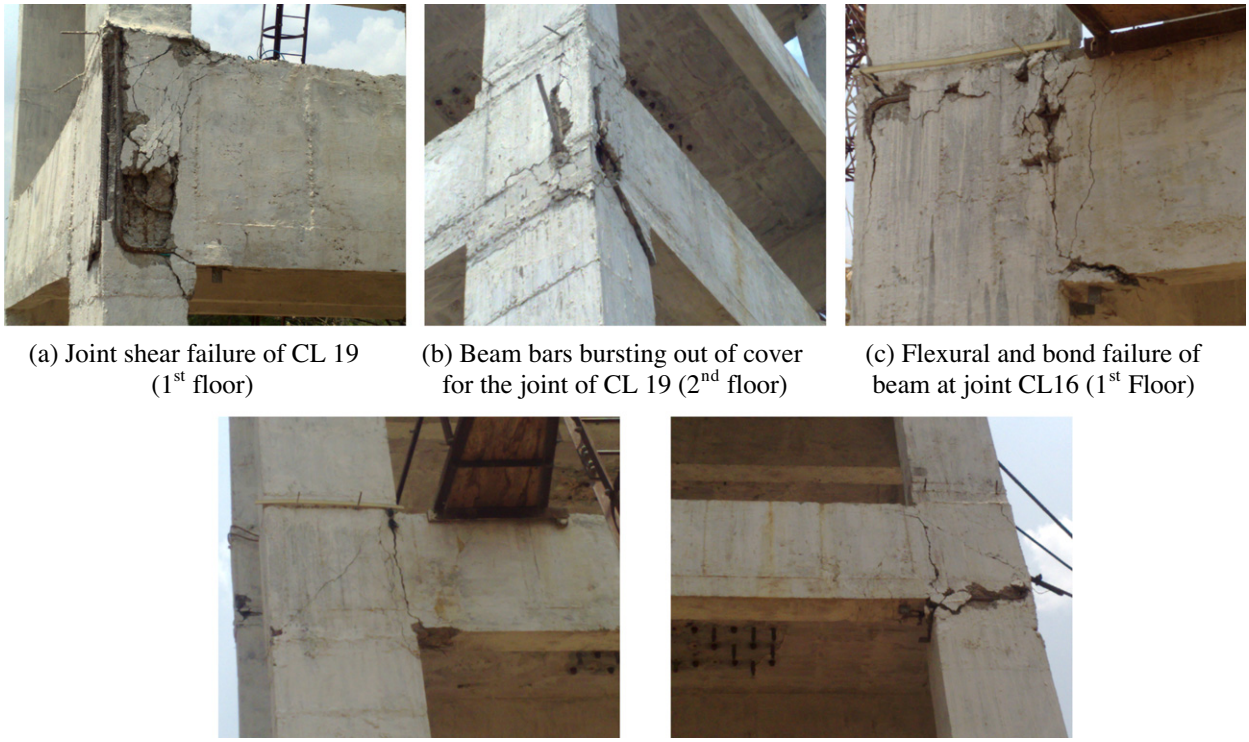
(a) CL 16 (Flexure-Compression) (b) CL 20 (Flexure-Compression)

(c) CL 15 (Flexure-Tension) (d) CL 19 (Flexure-Tension)

Fig. 8. Failure of columns at base under combined axial load and bending.

(a) Flexural failure of beam BF 205-1 (b) Torsional failure of beam BF223-1

Fig. 9. Failure modes observed in beams of the structure.



(a) Joint shear failure of CL 19 (1st floor)

(b) Beam bars bursting out of cover for the joint of CL 19 (2nd floor)

(c) Flexural and bond failure of beam at joint CL16 (1st Floor)

(d) Joint shear cracking and flexural cracking of beam at CL16 (2nd floor)

(e) Joint shear, beam-flexure and bond failure of beam bars at CL20 (2nd floor)

Fig. 10. Failure modes observed in beam-column joints of the structure.

was increased, the inter-storey drift increased and the structure entered the inelastic (nonlinear) range. It was observed that as the displacement increases, the contribution of relative displacement between third and fourth floor is smaller which is attributed to the joint failure at the third floor level.

5.1. Failure patterns

Figs 8–10 show various failure modes and patterns observed during the experiment. As the test started, the initial cracks were observed on the tension face at the base of the columns and at the tension face on beam ends for the beam BF 205–1 at first floor level. The corresponding base shear at this step was approximately 300 kN. In the next step, the beam BF205–2, i.e., at 2nd floor level started to display cracks and also the already formed cracks at the column bases and beams became wider. However, up to this point the beam–column joints had no signs of distress. At a base shear equal to 500 kN, the first shear cracks started to appear at the beam-column joints at 1st floor level. These cracks opened with the further increase in the load and more cracks at higher elevation (2nd and 3rd level) were also observed in the beams and beam-column joints. Additionally, at a base shear of approximately 500 kN, first cracks in the beam BF225–1 appeared at the ends of the beam near CL16 and CL20. Further increase in load led to significant widening of the existing cracks, spalling of concrete and formation of new cracks at upper floor levels. The failures at various locations in the structure at the peak load are described below.

Fig. 8a and b presents the failure of bottom storey columns on compression side, namely columns CL16 and CL20. The columns exhibited well-known failure modes of combined axial compression and bending. As the lateral load was applied and gradually in-

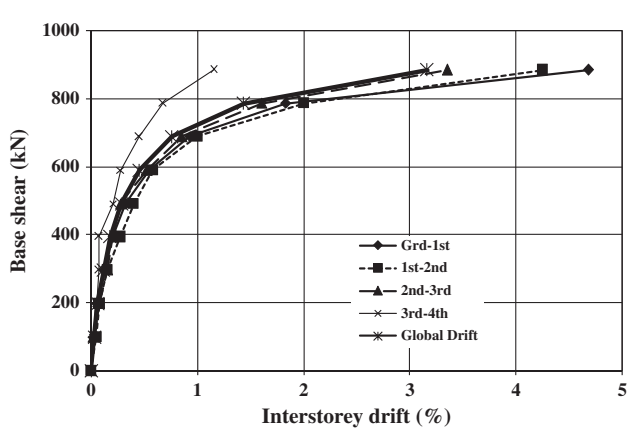


Fig. 11. Inter-storey drift as a function of base shear.

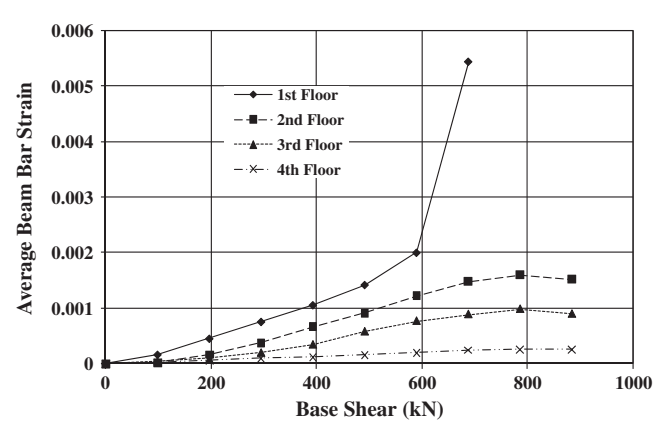


Fig. 12. Average beam re-bar strains for BF 205 at different floor levels.

creased on the structure, columns CL 16 and CL20 were exposed to increasing compressive forces combined with bending moment. Thus, due to this combined axial compression and bending, the column section started having tension cracks on the rear face. Further increase of the loads resulted in higher bending moments as well as axial forces on the column, and these tension cracks became bigger along the depth of the section due to the shifting of neutral axis towards the front face of the columns. Moreover, due to the shift of neutral axis, less area was available to resist higher compressive forces. Consequently, crushing of concrete on front face of the column occurred. The state of the columns at peak load is depicted in Fig. 8a and b.

Fig. 8 (c and d) illustrates the failure pattern of columns on tension side of CL15 and CL19, which were subjected to combined axial tension and bending moments. The columns were initially under compression due to self load of the building, but as the lateral load on the structure increased, the tensile forces on the two columns started to develop, along with bending moments. Under the action of the combined axial tension and bending moments, the columns started developing cracks from the rear face of the columns that propagated as the load increased, towards the front face of the columns. The spalling on the front face was nominal compared to that of CL16 and CL20 and major tension cracks were observed.

Fig. 9a depicts the failure mode of the beam BF 205, which is connected to CL15 at 1st floor, in flexural mode combined with bond slippage of the beam tension reinforcing bars. Due to lateral loading, bending moments were generated in the beam with hogging moments towards the end fixed with column CL16 and sagging moments towards the end fixed with column CL15. As a result, flexural tension cracks were observed initiating from the soffit of the beam and propagating towards the slab as shown in Fig. 9a. Due to high tensile stresses generated in the beam bottom bars, a slippage of the bars seems to have occurred. Spalling of concrete was observed on both the tension and compression faces of the beam due to extensive cracking and crushing, respectively.

Fig. 9b shows the failure of the beam BF 225, which is transverse to the direction of loading. As shown in Fig. 3, the load was applied on the structure through the slabs of each floor. As the lateral load increased, the beams transverse to the direction of loading in the front, namely BR21 and BF225, were pushed by the slab. This push was resisted by the stiffness provided at the ends due to restraining action of columns CL16 and CL20. Due to the end restraints, the beams suffered high compatibility torsion moments at the fixed ends. Nevertheless, this is not a typical seismic failure mode, since it can be attributed to the design of loading arrangement.

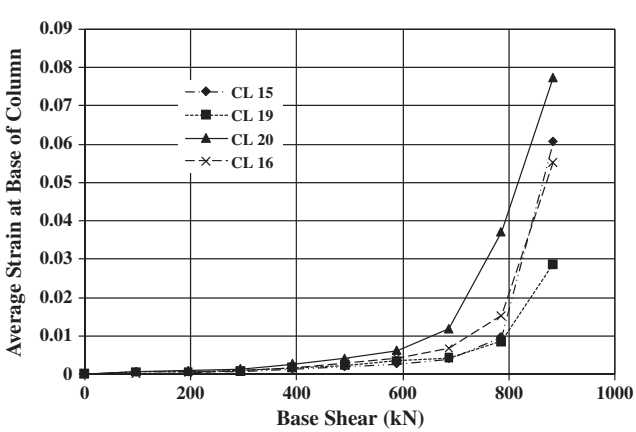


Fig. 13. Average surface strains at the base of the columns.

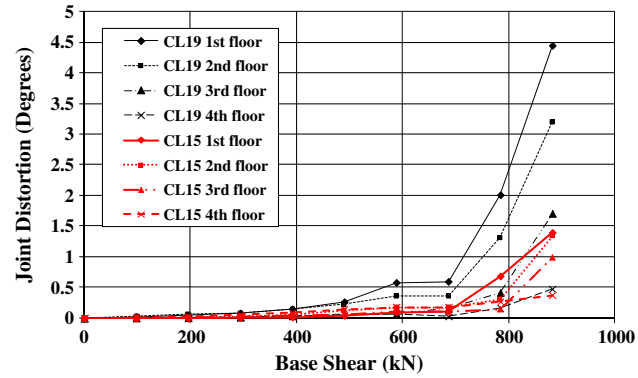


Fig. 14. Relative rotations of beams and columns framing into joints of CL 15 and CL 19.

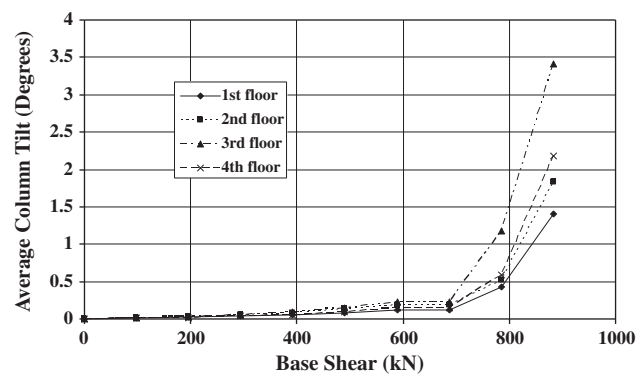


Fig. 15. Average tilt of the column of the structure.

Fig. 10 presents different types of joint failures observed in the structure. Under the action of lateral forces, beam-column joints are subjected to large shear stresses in their core. Typically, high bond stress requirements are also imposed on reinforcement bars passing through the joint. The axial and joint shear stresses result in principal tension and compression that leads to diagonal cracking and/or crushing of concrete in the joint core. The flexural forces from the beams and columns cause tension or compression forces in the longitudinal reinforcements passing through the joint. During plastic hinge formation, relatively large tensile forces are transferred through bond. When the longitudinal bars at the joint face are stressed beyond yield, splitting cracks are initiated along the bar at the joint face. If the concrete cover of the reinforcement bars is inadequate and if the joint core is not confined by confining reinforcement in the form of stirrups, the cover concrete is spalled off due to the pressure exerted by the beam reinforcement bars. Most severe joint failures were found in the case of column CL 19. This might be attributed to the relatively low column depth (400 mm) compared to beam depth (1000 mm). In such cases, plasticization of columns can occur, which may also lead to damage ingress in the joint core. Moreover, there was high eccentricity between the beam and the column, since the beam of width 230 mm was joined at the face of the column having width of 900 mm.

Fig. 10a illustrates the failure of joint of CL 19 at first floor. High stresses in the joint resulted in diagonal cracks in the core, followed by cover spalling due to the pressure exerted by the beam longitudinal reinforcement. Fig. 10b shows the failure of joint of CL 19 at 2nd floor level, which shows the beam bar bursting out of the joint. This is a typical failure mode for joints with unrestrained bars. This occurred since in order to provide the development length of the beam main reinforcement, the bent bars had a

Table 4
Summary of assumptions and modeling aspects considered by the participants of round robin exercise.

No.	Software	Concrete const. law	Rebar const. law	Modeling of Joints	Flexure hinge	Shear Hinge	Torsion Hinge	Axial-Moment Interaction	Geometric Nonlinearity	Slab Modeling
1	SAP2000	Mander	Strain hardening	Rigid	Yes	No	No	Yes	No	Only mass
2	SAP2000	Mander	IS 456	Rigid	Yes	No	No	Yes	No	Rigid Diaphragm
3	Ansys	IS456	Strain hardening	Rigid	Yes	No	No	Yes	Yes	Rigid Diaphragm
4	SAP2000	Hinges by FEMA356	Hinges by FEMA356	Rigid	Yes	No	No	No	No	Only mass
5	SAP2000	SAP Default hinges	SAP Default hinges	Rigid	Yes	No	No	No	No	Not modeled
6	SAP2000	Kent and Park	Strain hardening	Rigid end offsets	Yes	Yes	No	Yes	Yes	Shell Elements
7	SAP2000	Hinges by FEMA356	Hinges by FEMA356	Rigid end offsets	Yes	Yes	No	Yes	No	Mass + Rigid Diaphragm
8	SAP2000	Mander	Strain hardening	Rigid	Yes	No	No	No	No	Shell Elements
9	Ansys	IS 456	IS 456	Rigid end offsets	Yes	No	No	30% axial load considered	No	Shell elements
10	SAP2000	IS 456	IS 456	Rigid	Yes	No	No	Yes	No	Rigid Diaphragm

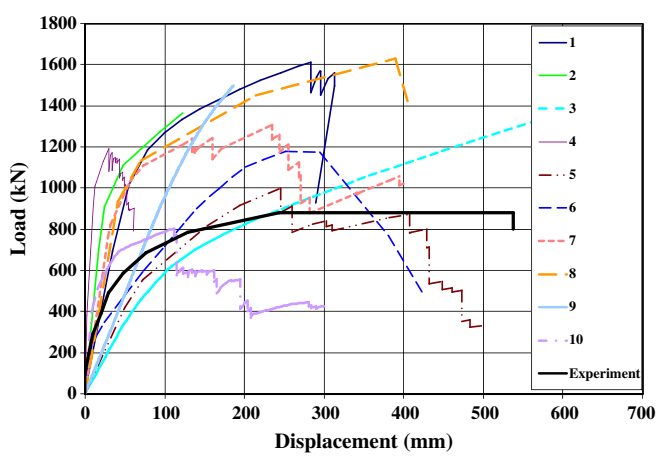


Fig. 16. Summary of analysis results submitted by participants of round robin exercise.

long free length beyond the bending length and there were no transverse reinforcement to provide any restrain at this region. Such a failure can, in general, be prevented if proper confining reinforcement is provided in the joint core. Fig. 10c depicts the failure of the joint of CL16 at first floor level that exhibited bond failure along with beam flexural failure and spalling of side cover due to pressure exerted by the reinforcement. High tension force in the beam reinforcement resulted in bond deterioration and ultimately failure with splitting of concrete. Furthermore, large cracks along with spalling of concrete can be observed at the beam-column interface. Fig. 10d shows a typical diagonal (shear) crack in the joint of CL16, 2nd floor with flexural cracks in the beam. The diagonal cracks in the joints are formed due to principal tensile stresses generated as a result of axial and joint shear stresses. As the lateral forces were increased on the structure, the joint shear stress increased and in combination with the axial stresses, resulted in diagonal tension that was responsible for the development of diagonal tension cracks. Fig. 10e presents diagonal shear crack in the joint of CL20, 2nd floor during the test with flexural cracks in the beam and bond failure of the tension reinforcement. It can be observed that a clear diagonal shear crack appeared in the joint during the test, but it was not further opened and essentially the failure was transferred through bond mechanism. Although the beam longitudinal reinforcement was bent up to the required development length inside the column, it appears that such bending may not be adequate to prevent bond failure.

5.2. Inter-storey drifts

Fig. 11 illustrates the inter-storey drifts between ground to 1st floor, 1st to 2nd floor and so on, as a function of base shear on CL 16 side. Furthermore, in the same plot, the roof drift obtained as the lateral roof deflection divided by the total height of the structure is given. Maximum inter-storey drifts were obtained between the ground to first floor and first to second floor and were of the order of 4.5%. Drifts between second to third floor was equal to 3.5%, which was also the order of global drift. The inter-storey drifts between the third and roof level were in the range of 1–1.5%, which explains why greater damage levels were concentrated within lower floors.

5.3. Strain data

The average beam bar strains of beams BF 205 are plotted in Fig. 12 for 1st, 2nd, 3rd and 4th floor levels of the structure. The plot clearly shows that as the base shear increased, the average beam bar strain increased almost linearly up to a base shear value of 500 kN and thereafter going in the non-linear range. As expected, the maximum strains were obtained at 1st floor level and the strain gauges at that level broke after the yielding of the rein-

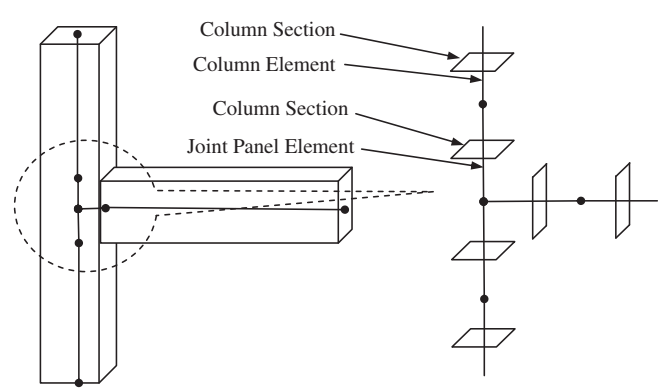


Fig. 17. Modeling of joint panel.

forcement bars. These strain gauges could read only up to a base shear of 700 kN. The strain gauges at other floors did not show very high values, which is mainly due to the fact that the failure modes at second and third floors were mainly governed by the bond failure of beam bars and the shear failure of the joint. The fourth floor beams were almost undamaged. As discussed earlier, the dial gauges were installed on the tension face of the columns at the base. The dial gauges read the total extension over a gauge length, which was then converted to average surface strains at the base of the columns and plotted in Fig. 13. It can be observed that the average strains grow linearly with base shear up to a base shear of approximately 400–500 kN, and thereafter start to increase at a higher rate. The rate of increase of strain becomes very high after a base shear of 700 kN. As expected and verified by failure modes, column CL 19 has minimum strains.

5.4. Rotations

The tilt meters were used in order to get the information on the rotation of members and joints. The relative rotation between beams and columns framing into the joints of CL15 and CL19 for different floors are given in Fig. 14. Since the tilt meters were placed very close to the joint faces on beams and columns, this relative rotation is also a measure of rotation (shear distortion) of the joint. It is clear that the relative rotation is higher for the joints of CL19, which is attributed to the low depth of the column as compared to that for the joints of CL15. Fig. 15 shows the average tilt of the columns at various floors. In addition, the tilt increased till third floor and reduced in the case of fourth floor, which is also easily detected in the deflection profile of the structure.

6. Round robin exercise

The experiment was conducted as a round robin exercise where various participants from different academic and research institutes had participated and presented their results. A summary of the approaches followed by a comparison of the results in the form of base-shear vs. roof deflection curves are given in Table 4 and Fig. 16, respectively. All participants used the conventional non-adaptive pushover method and modeled the structure using frame elements, which is reasonable since modeling of the whole structure using 3D solid elements and discrete bar elements is unnecessarily time consuming. Most of the participants modeled the joints as rigid points, while others took into account finite dimensions of the joints, by using rigid end offsets. However, a more elaborate consideration of joint distortion was not presented in any of the models. All participants modeled the formation of flexural hinges, while six of them also considered axial force-moment interaction. Four participants considered the effect of confinement in the concrete model. These two aspects are quite important while performing the inelastic analysis and neglecting them may lead to quite misleading results in terms of both load and displacement estimates. This is also proven by the results of participants 4 and 9 who neglected both aspects and got worse results.

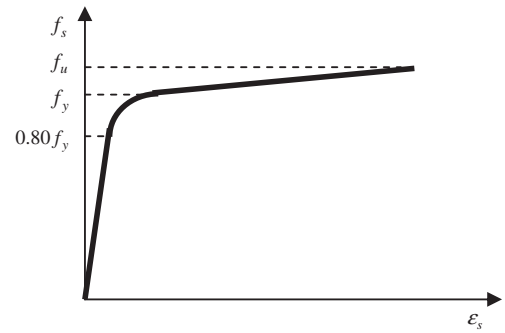
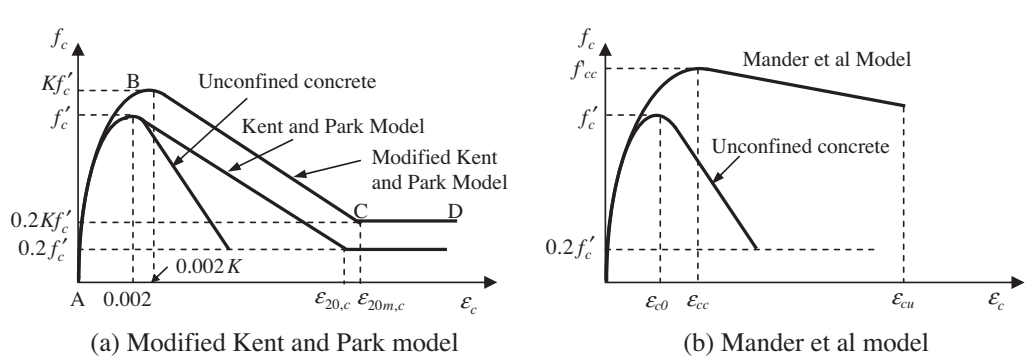


Fig. 19. Theoretical stress–strain curves for reinforcing steel used in this work [31].

Shear hinge was modeled only by two participants, while torsion hinge was not modeled by anyone. Modeling of slab in each floor was done in different ways: (i) not modeling at all, (ii) considering only the effect of weight of the slab, (iii) modeling slab as rigid diaphragm, and (iv) modeling as shell elements. However, none of the participants modeled any nonlinearity in the slab or slab-beam intersection. All participants used SAP2000® software except participant 3 and 9 who used Ansys® software to perform the analysis. From the aforementioned discussion, it is obvious that almost every participant followed a different approach to model the structure. Even while modeling common parameters, e.g., flexural hinge, there were several differences among the participants, such as different constitutive relationships for concrete and reinforcement; deriving moment–curvature relation using equivalent rectangular stress block approach or fiber approach; using different formulations for plastic hinge lengths, etc. As can be expected, based on the assumptions of the various approaches, participants submitted quite widespread results with expected base shear values ranging from 800 kN to 1600 kN and the roof displacement values ranging from 60 mm to 600 mm. As expected the variation in displacement prediction is much higher as compared to base shear prediction since the models for the determination of load carrying capacity are better understood.

As shown in Fig. 16, none of the analytical results matched satisfactorily the experiment ones. Although, curve 5 seems to be close to the experimental one, it may not be considered to be a suitable model since it was obtained using the default models of SAP2000® software and it is more a matter of chance than technical suitability that the results are relatively close to those of the experiment. In contrast, except from the ductility results, curve of participant 10 is quite acceptable both for the initial stiffness



(a) Modified Kent and Park model (b) Mander et al model

Fig. 18. Theoretical stress–strain curves for confined and unconfined concrete.

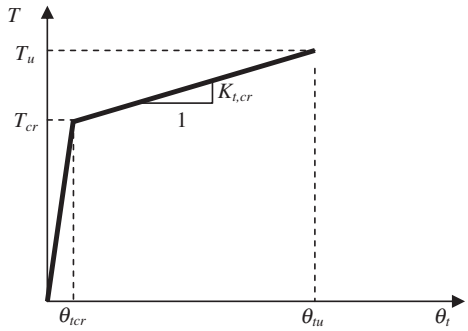


Fig. 20. Typical torsion hinge characteristics for the section [5].

and the base shear capacity. Therefore, it appears that model 10 could have produced good results if just the confinement effect would have been modeled, but it cannot be commented any further due to lack of related data. Considering the actual failures occurred in the structure, it can be said that the joint inelastic behavior contributed significantly in structural response. This seems to be one of the major reasons for the mismatch between the experiment and pre-test analytical results, since no participant considered the inelastic behavior of the beam-column connections. It would have been really valuable if the blind numerical prediction of the observed response involved more refined means to model the non-linear static response of the structure, such as adaptive methods, fiber analysis-based software with advanced simulation capabilities, like OpenSees [34] etc.

7. Post-test analysis

7.1. Modeling of frame members, joints and slabs

Given the good constitutive laws and crack propagation criteria, modeling of members and joints using 3D solid elements and discrete bar elements with bond model generally provide results close to reality [35]. However, on the other hand, it is understood that such a model is unreasonable to use in practice due to excessive modeling time and computational costs involved. Therefore, in order to provide a solution that is simple enough for usage in practice while being reasonably accurate, the beams and columns were modeled as 3D beam (frame) elements, with six degrees of freedom at both nodes. Frame members are modeled as line elements connected at points (joints). The slabs were modeled using four-noded quadrilateral shell elements. To consider the finite dimensions of the joints, each of them was modeled by dividing the frame elements into two frame elements, one to represent beam or column element and one to represent joint panel. This was also needed to provide joint spring characteristics to consider joint distortion as will be explained later. The modeling of joint panel is explained in Fig. 17.

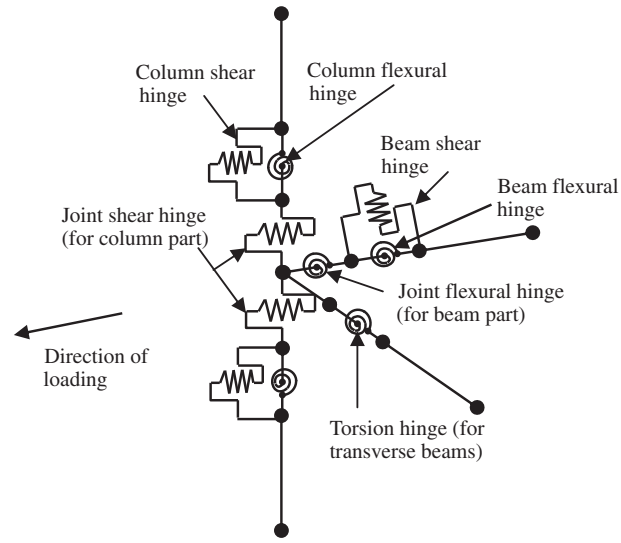


Fig. 22. Hinges assigned to the members and core of a typical joint.

7.2. Modeling of nonlinearities

7.2.1. Flexural hinge

The stress–strain characteristics of concrete confined by transverse reinforcement exhibits a more ductile behavior than its unconfined counterpart [5,6,36,37]. Therefore, in order to generate moment-rotation characteristics for a section, the first step is to obtain the stress–strain curve for the confined concrete. Many researchers have proposed models to estimate the stress–strain curve for the confined concrete over the last decades [36–40] to late twentieth and early twenty first century [41–44]. Many other models can be found in literature. However, among these models, the modified Kent and Park model [37] (Fig. 18a) and the Mander model [42] (Fig. 18b) are more popular, mainly because they offer a good balance between simplicity and accuracy. In this work, the modified Kent and Park model [37] (Fig. 18a) was followed, however, the authors believe that the Mander model [42] would also provide similar results. The stress–strain characteristics for the reinforcement steel used in this work is considered by suitably modifying the curve suggested by Indian code [33] to include strain hardening in the post yield portion of the curve (Fig. 19). Same curve was followed for reinforcement bars in tension and compression. Once the stress–strain curves for steel and concrete are formulated, the moment–curvature characteristics of the section were derived using the standard procedure considering the

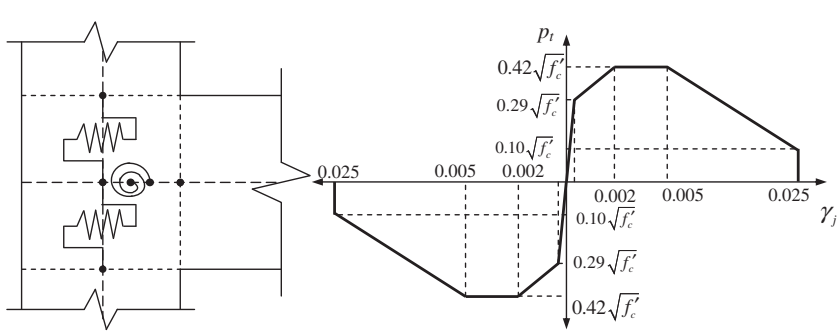


Fig. 21. Principal tensile stress–shear deformation relations used for the joint [43,44].

equilibrium of forces and compatibility of strains. The generated moment–curvature characteristics were converted to moment-rotation characteristics using the following expressions for yield and ultimate rotations:

$$\theta_y = \int_0^L \varphi_y dx = \int_0^L \frac{M_y}{EI} dx \tag{3}$$

$$\theta_u = \theta_y + (\varphi_u - \varphi_y) l_p \tag{4}$$

where, l_p is the plastic hinge length, which was calculated using the formulation suggested by Baker for confined concrete [5,39]. Alternatively, the expression suggested by Pauley and Priestley [6] may also be used. However, for typical beam and column dimensions, a value of l_p equal to half of the effective depth of the section can be used with sufficient accuracy.

7.2.2. Shear hinge

To predict the shear force–deformation characteristics, an incremental analytical approach was followed [8], which is based on the truss mechanism. In this model, the stirrup strain is gradually increased with a small increment and the resisting shear at each step is calculated. The stress state is characterized by a biaxial stress field in the concrete and a uniaxial tension field in the shear reinforcement. Moreover the theoretical basis given by Kupfer and Bulicek [45] for the equilibrium condition of stresses and compatibility condition of strains for the concrete element shown is followed. The equilibrium condition of stresses, compatibility condition of strains and constitutive laws are then used to obtain the complete shear force vs. deformation characteristics for the members. The method is straightforward and easily programmable. However, a detailed description of the approach is beyond the scope of this paper and details of the model can be found in [8].

7.2.3. Torsional hinge

The torsional hinge characteristics of the section were determined on the basis of Space Truss analogy [5]. The cracking torsion, T_{cr}, is calculated as follows:

$$T_{cr} = 0.33\sqrt{f'_c}(A_c^2/P_c) \tag{5}$$

where f'_c = standard cylinder compressive strength of concrete, considered as 0.8 times the standard cube strength of concrete, A_c = gross area of concrete section in mm^2, P_c = perimeter of concrete section in mm.

The ultimate torsional resistance, T_u, of the section is calculated from:

$$T_u = 2A_o A_{sv} f_{sv} Cot\theta / s_v \tag{6}$$

in which, A_o = gross area enclosed by shear flow path, considered as 0.85 times the area enclosed by the centerline of the outermost closed transverse reinforcement, A_{sv} = area of one leg of transverse reinforcement, f_{sv} = yield/ultimate stress of transverse reinforcement, s_v = centre to centre spacing of transverse reinforcement.

The cracked stiffness of the section, $K_{t,cr}$ is given by [5]:

$$K_{t,cr} = E_s(B_o D_o)^2 A_{sv}\sqrt{m_t}/l\{(B_o + D_o)s_v) \tag{7}$$

where, B_o = shorter dimension of transverse reinforcement, D_o = longer dimension of transverse reinforcement, E_s = modulus of elasticity of transverse reinforcing steel, M_t = ratio of yield stress of transverse reinforcement to that of longitudinal reinforcement, l = span length.

Typical torsional hinge characteristics are shown in Fig. 20.

7.2.4. Joint hinge

Since the structure suffered severe damages in the joint regions, it was very important to model the nonlinearities in the beam-column joints in order to capture their real behavior. In addition, it is true that a detailed modeling of the structure using 3D solid elements for concrete with an associated constitutive law, such as microplane model, with reinforcement modeled as bar elements and a specified bond characteristics is likely to give realistic results [35]. However, such an analysis is extremely time consuming and therefore is discouraging for practitioners. Therefore, in this work, a new joint model proposed by Sharma et al. [38] is followed, which has been shown to be quite effective in capturing realistically the response of poorly detailed beam-column joints. The model uses limiting principal tensile stress in the joint as the failure criterion so that due consideration is given to the axial load on the column. The spring characteristics are based on the actual deformations taking place in the sub-assemblage due to joint shear distortion [46].

For an exterior joint, two shear springs and one rotational spring were used to model the joint distortion (Fig. 21). For beam-column joints with the beam bars bent in the joint, the curve for principal tensile stress vs. shear strain that was validated and used in this work is shown in Fig. 21. This curve is based on the recommendations of Priestley [47]. The rotational and shear spring characteristics are derived using the relation shown in Fig. 21 and equilibrium criteria for the joints. The complete details of this process are given in reference [46].

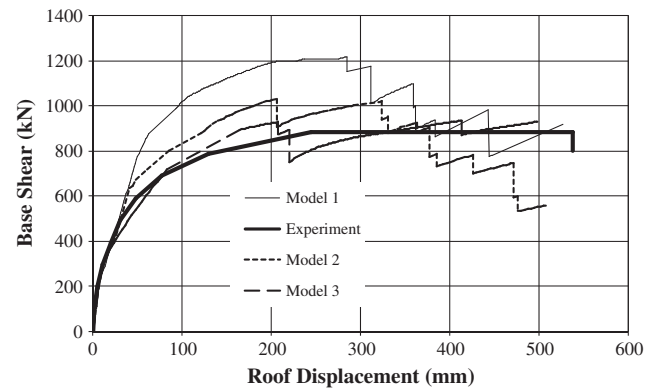


Fig. 23. Comparison of results.

7.3. Computational details

The hinge characteristics, once obtained, were assigned to the frame members. The hinges assigned on a typical joint of the structure in the program and their physical significance is displayed in

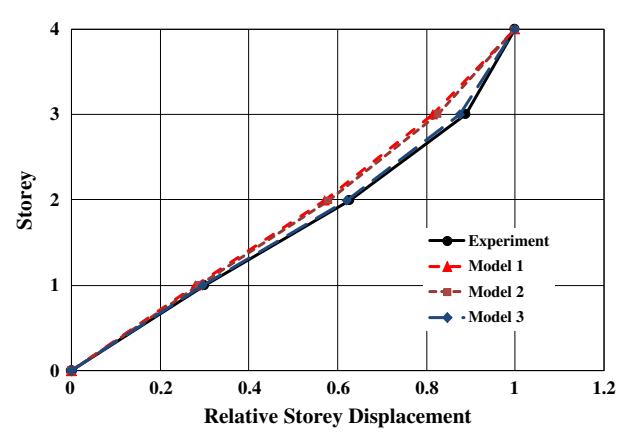


Fig. 24. Comparison of deflected shape of the structure.

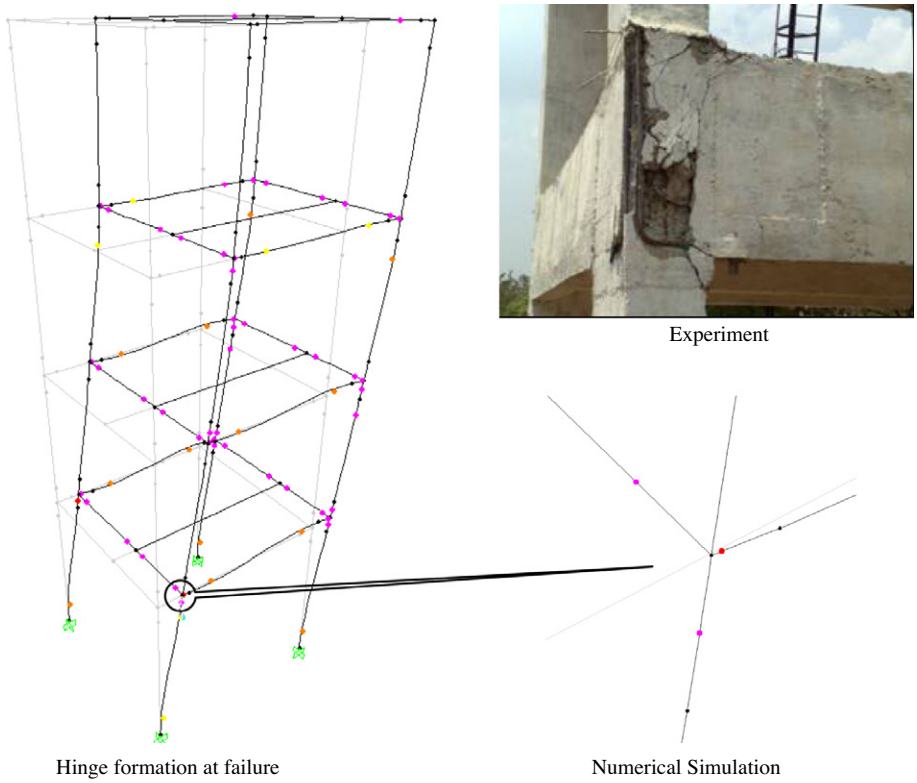


Fig. 25. Failure mode of the structure with emphasis on joint of CL 19 at 1st floor level.

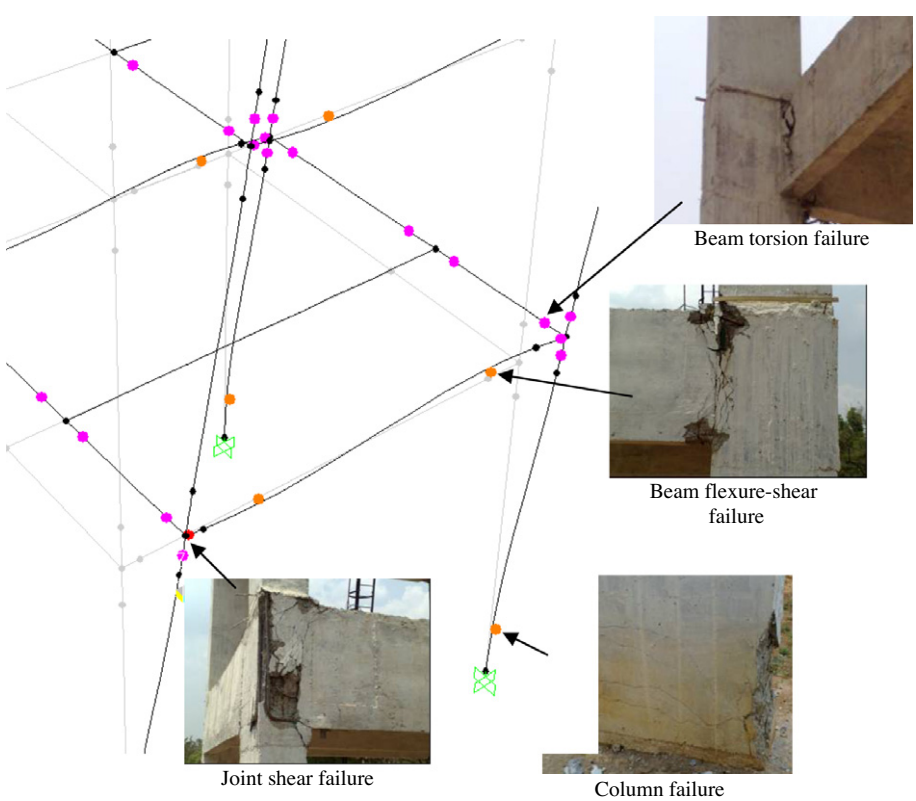


Fig. 26. Comparison of failure modes as experimentally and numerically derived.

Fig. 22. Since the loading was uni-directional, no joint hinges were provided for the transverse beams of the joints. All hinges were zero length springs, including joint hinges. Modeling was done utilizing the capabilities of the commercial software SAP2000. The modulus of elasticity, E_c, was equal to $4730(f'_c)^{0.5}$ [48], while the cracked stiffness was considered by using modulus of elasticity

$0.5E_c$ [4]. The effect of confinement and axial forces was considered while deriving the flexural hinge characteristics.

7.4. Numerical results

In order to have a comparison among modeling techniques, three cases were analyzed, with different types of nonlinear hinges models:

1. Model 1, with flexural and shear hinges only;
2. Model 2, with torsional hinges along with flexural and shear hinges;
3. Model 3, with joint characteristics along with torsional, flexural and shear hinges.

Fig. 23 shows the comparison of experimental and analytical results for the examined cases. It can be observed that the first model over-predicts the strength of the structure. However, the initial stiffness obtained from the analysis in this case is quite close to the experimentally obtained one. The over-prediction of strength was expected, since the analysis considered only moment and shear hinges, whereas in the experiment it was found that the torsional and joint failure were also dominant. After considering the torsional effects, the predicted maximum base shear approximates better the experimentally obtained value. However, the predicted maximum base shear is still higher than the actual base shear. This is attributed to the fact that in this case the nonlinear characteristics of the joints were not modeled. Finally, after considering the joint characteristics, torsional effects, moment and shear characteristics the analysis using third model predicted very well the load-deformation behavior of the structure. The numerical results follow the experimental ones very closely. It has to be noted that the geometric nonlinearity in terms of P-delta effects was considered in all models and no calibration was performed to obtain the presented results.

Fig. 24 presents a comparison of the experimentally observed and numerically simulated deflected shape of the structure for each analysis case, with respect to the point when the structure reaches the first peak. Since the computational models and the experimental setup reach peak base shear at different displacement, for better comparison of the deflected shape, the actual values of the storey displacement were normalized with respect to roof displacement. It can be seen that the numerically obtained displacement shape for Models 1 and 2 display a parabolic shape for the structure and do not match the experimentally observed profile. This discrepancy is attributed to the rigid behavior of the joints. However, in the experiment, due to the failure at joint levels, the displacement of the roof level was much less than would be expected in the case of shear building behavior. In order to simulate this phenomenon, modeling of joint nonlinearities becomes extremely important and therefore explains why the deflected shape obtained from Model 3 matches closely the experimentally observed one. Thus, it can be concluded that the third model could simulate almost all types of failure modes that were observed in the experiment, since not only the base shear, but also the deflected shape of the structure could be successfully captured.

Fig. 25 depicts the various hinges formed in the structure in the computational model with flexural, shear, torsional and joint hinges. A zoomed view of joint at 1st floor level for column CL19 is provided to illustrate how the model is able to capture the real behavior of the joint. Similarly, an enlarged view of the first floor of the structural model is shown in Fig. 26 where each hinge and its corresponding physical significance in real life case are shown. The consistency between the hinges obtained in the analysis and the failures in the experiment is remarkable.

The same modeling approach was earlier utilized by the authors [49] to predict the non-linear static response of a small-scale RC frame structure and a good agreement with experimental results was obtained in that case as well. However, as mentioned earlier, probably the numerical prediction based on more refined means to model the non-linear static response of the structure, such as fiber analysis could also have provided good results.

8. Summary and conclusions

In this study, a full scale experiment was conducted on a RC frame which was a replicate of a substructure of an existing office building in India. The structure was constructed with non-seismic detailing and the foundation was constructed with rock anchors to avoid any possible rotation during the experiment. The failure patterns displayed the vulnerability of RC buildings with non-conforming detailing which tend to fail in undesirable failure mechanisms, such as joint shear failures and bond failures.

Moreover, the experiment was carried out as a round robin exercise and various institutes in India presented the results in the form of pushover curves using various approaches. A large dispersion in the results was observed. As part of a post-test analysis three different modeling options were considered:

1. modeling moment and shear hinges in the members only (Model 1),
2. modeling of torsional hinges along with the moment and shear hinges (Model 2), and
3. modeling of joint behavior along with torsional, moment and shear hinges (Model 3).

It has been shown that in order to capture the overall behavior of RC structures, neglecting the inelasticity in the joints can lead to inaccurate results. The first two models over-predicted the base shear resistance of the structure and inaccurate deflected shapes were also derived. In contrast, it was found that via Model 3, not only the pushover curves, but also the deflected shape of the structure as well as the failure modes and locations could be satisfactorily simulated. Similar trends were observed in a similar study by the authors, in which they implemented the aforementioned modeling options for the dynamic non-linear analysis of RC frame structures [49].

Acknowledgements

The experiment was carried out at Central Power Research Institute (CPRI), Bangalore under the research project funded by Bhabha Atomic Research Centre (BARC), Mumbai. The experiment would not have been successful without the untiring efforts of Mr. D. Revanna, Mr. M.N. Gundu Rao, Mr. B.N. Dinesh Kumar and Dr. R. Ramesh Babu of CPRI. The authors are also highly thankful to Mr. R.V. Nandanwar, Mr. S.N. Bodele and Mr. M.A. Khan, of Reactor Safety Division, BARC and Mr. Philip, Mr. Ashok Kumar and Mr. Rajan of Earthquake Engineering and Vibration Research Centre, CPRI for their valuable support.

References

[1] Applied Technology Council. Seismic evaluation and retrofit of concrete buildings. Report No. ATC-40, Applied Technology Council, California; 1996.
[2] Building Seismic Safety Council (BSSC). NEHRP guidelines for the seismic rehabilitation of buildings. Report FEMA-273 (Guidelines) and Report FEMA-274 (Commentary), Washington (DC); 1997.
[3] Federal Emergency Management Agency. Pre-standard and Commentary for Seismic Rehabilitation of Buildings. Report No. FEMA-356, Washington (DC); 2000.
[4] Applied Technology Council. Improvement of nonlinear static seismic analysis procedures. Report No. FEMA-440, Washington (DC); 2005.

[5] Park R, Paulay T. Reinforced concrete structures. New York: John Wiley & Sons; 1975.
[6] Paulay T, Priestley MJN. Seismic design of reinforced concrete and masonry buildings. New York: John Wiley publications; 1992.
[7] Samra RM. Ductility analysis of confined columns. J Struct Eng ASCE 1990;116(11):3148–61.
[8] Watanabe F, Lee JY. Theoretical prediction of shear strength and failure mode of reinforced concrete beams. ACI Struct J 1998;95(6):749–57.
[9] Biddah A, Ghobarah A. Modelling of shear deformation and bond slip in reinforced concrete joints. Struct Eng Mech 1999;7(4):413–32.
[10] Ziyaeifara M, Noguchib H. A refined model for beam elements and beam-column joints. Comp Struct 2000;76(4):551–64.
[11] Cosenza E, Manfredi G, Verderame GM. A fibre model for push-over analysis of under-designed reinforced concrete frames. Comp Struct 2006;84:904–16.
[12] Ghobarah A, Biddah A. Dynamic analysis of reinforced concrete frames including joint shear deformation. Eng Struct 1999;21:971–87.
[13] Fabbrocino G, Verderame GM, Manfredi G, Cosenza E. Structural models of critical regions in old-type R.C. frames with smooth rebars. Eng Struct 2004;26:2137–48.
[14] Kaminosono T. The full-scale seismic experiment of a seven-story reinforced concrete building, Part 1: Outline of test results. In: Proceedings of the 6th Japan earthquake engineering, symposium; 1982. P. 865–72.
[15] Pinho R, Elnashai AS. Dynamic collapse testing of a full-scale four storey RC frame. ISET J Earthq Tech 2000;37(4):143–63.
[16] Jeong SH, Elnashai AS. Analytical assessment of an irregular RC full scale 3D test structure. Mid-America Earthquake Center report for National Science Foundation Grant EEC-9701785; 2004.
[17] Weng YT, Lin KC, Hwang SJ. Experimental and analytical performance assessment of in-situ pushover tests of school buildings in Taiwan. In: 4th International Conference on Earthquake Engineering; 2006 [Paper No. 154].
[18] Tu YH, Jiang WC, Hwang, SJ. In situ pushover test of a school building in Taiwan. NCREE Newslett. 2006:1(1).
[19] Calvi GM, Magenes G, Pampanin S. Experimental test on a three storey R.C. frame designed for gravity only. In: 12th European Conference on Earthquake Engineering; 2002 [Paper reference 727].
[20] Chopra AK, Goel RK. A modal pushover analysis procedure for estimating seismic demands for buildings. Earthq Eng Struct Dyn 2002;31:561–82.
[21] Chopra AK, Goel RK. A modal pushover analysis procedure to estimate seismic demands for unsymmetric-plan buildings. Earthq Eng Struct Dyn 2004;33:903–27.
[22] Isakovic T, Fischinger M. Higher modes in simplified inelastic seismic analysis of single column bent viaducts. Earthq Eng Struct Dyn 2006;35:95–114.
[23] Paraskeva TS, Kappos AJ, Sextos AG. Extension of modal pushover analysis to seismic assessment of bridges. Earthq Eng Struct Dyn 2006;35:1269–93.
[24] Kunnath SK. Identification of modal combination for nonlinear static analysis of building structures. Computer-Aided Civ Infrastruct Eng 2004;19:246–59.
[25] Han SW, Chopra AK. Approximate incremental dynamic analysis using the modal pushover analysis procedure. Earthq Eng Struct Dyn 2006;35:1853–73.
[26] Fajfar P, Gasperic P. The N2 method for the seismic damage analysis of RC buildings. Earthq Eng Struct Dyn 1996;25:31–46.
[27] Kilar V, Fajfar P. Simplified push-over analysis of building structure. In: Proceedings of the 11th World Conf on Earthq Eng; 1996 [Paper 1011].
[28] Kilar V, Fajfar P. Simple push-over analysis of asymmetric buildings. Earthq Eng Struct Dyn 1997;26:233–49.
[29] Vamvatsikos D, Cornell AC. Incremental dynamic analysis. Earthq Eng Struct Dyn 2002;31(3):491–514.
[30] Nicknam A, Ahmadi HR, Mahdavi N. A comparative study of the traditional performance and the incremental dynamic analysis approaches (IDA). In: Proceedings of the 14th world conference on, earthquake engineering; 2008.
[31] IS 1786: 1985. Indian Standard, Specification for high strength deformed steel bars and wires for concrete reinforcement (third revision). New Delhi: Bureau of Indian Standards.
[32] IS 13920: 1993. Indian Standard, Ductile detailing of reinforced concrete structures subjected to seismic forces - code of practice. New Delhi: Bureau of Indian Standards.
[33] IS 456:2000. Indian Standard plain and reinforced concrete – code of practice. New Delhi: Bureau of Indian Standards.
[34] Mazonni S, McKenna F, Scott MH, Fenves GL. OpenSees user command language manual; 2007. <http://opensees.berkeley.edu/OpenSees/manuals/usermanual/>.
[35] Sharma A, Genesio G, Reddy GR, Eligehausen R. Nonlinear dynamic analysis using microplane model for concrete and bond slip model for prediction of behavior of non-seismically detailed RCC beam–column joints. J Struct Eng, SERC 2009;36(4):250–7.
[36] Kent DC, Park R. Flexural members with confined concrete. J Struct Div, ASCE 1971;97:1969–90.
[37] Park R, Priestley MJN, Gill WD. Ductility of square-confined concrete columns. J Struct Eng, ASCE 1982;108:929–50.
[38] Chan WL. The ultimate strength and deformation of plastic hinges in reinforced concrete frameworks. Mag Conc Res 1955;7(21):121–32.
[39] Baker ALL, Amarakone AMN. Inelastic hyperstatic frames analysis. In: Proceedings of the international symposium on the flexural mechanics of reinforced, concrete; 1964. p. 85–142.
[40] Sargin M, Ghosh SK, Handa VK. Effects of lateral reinforcement upon the strength and deformation properties of concrete. Mag Conc Res 1971;23:99–110.
[41] Sheikh SA, Uzumeri SM. Analytical model for concrete confinement in tied columns. J Struct Div, ASCE 1982;108:2703–22.
[42] Mander JB, Priestley MJN, Park R. Theoretical stress–strain behavior of confined concrete. J Struct Eng, ASCE 1988;114(8):1804–26.
[43] Rueda JEM, Elnashai AS. Confined concrete model under cyclic load. Mater Struct 1997;30(3):139–47.
[44] Li YF, Chen SH, Chang KC, Liu KY. A constitutive model of concrete confined by steel reinforcements and steel jackets. Can J Civ Eng 2005;32:279–88.
[45] Kupfer H, Bulicek H. A consistent model for the design of shear reinforcement in slender beams with I- or Box-shaped cross section. In: Proceedings of international workshop on concrete shear in, earthquake; 1991. p. 256–65.
[46] Sharma A, Eligehausen R, Reddy GR. A new model to simulate joint shear behavior of poorly detailed beam–column connections in RC structures under seismic loads, part I: exterior joints. Eng Struct 2011;33:1034–51.
[47] Priestley MJN. Displacement based seismic assessment of reinforced concrete buildings. J Earthq Eng 1997;1(1):157–92.
[48] ACI 318. Building code requirements for structural concrete (ACI 318–08) and commentary (ACI 318R–08)". Detroit: American Concrete Institute; 2008.
[49] Sharma A, Reddy GR, Eligehausen R, Vaze KK. Experimental and analytical investigations on seismic behavior of reinforced concrete framed structure by pushover method. Struct Eng Mech 2011;39(1):125–45.